Experiment No:16
Experiment to understand EDA-Quantitative and Qualitative analysis.

AIM:
To understand EDA-Quantitative and Qualitative analysis.

DESCRIPTION:
Understand the importance of EDA-Quantitative and Qualitative analysis.

CODE:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
tips=sns.load_dataset('tips')
print(tips.head())
print("\n\n")

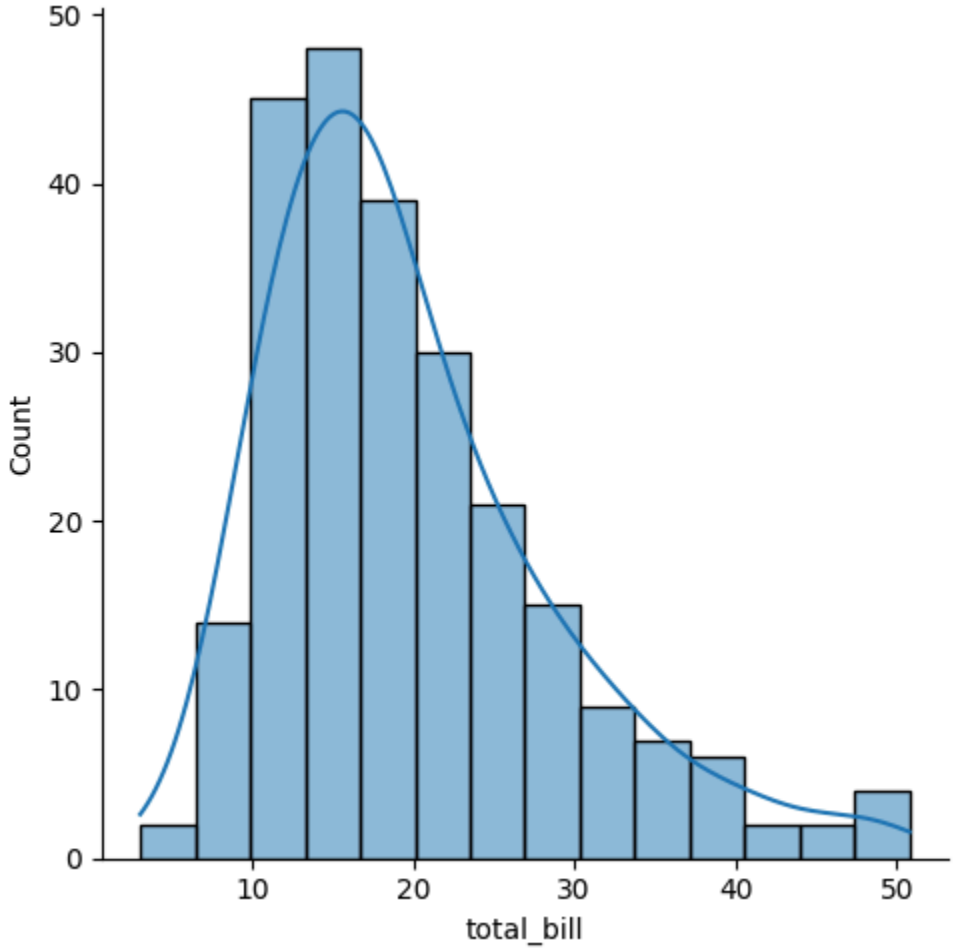

sns.displot(tips.total_bill,kde=True)
plt.show()

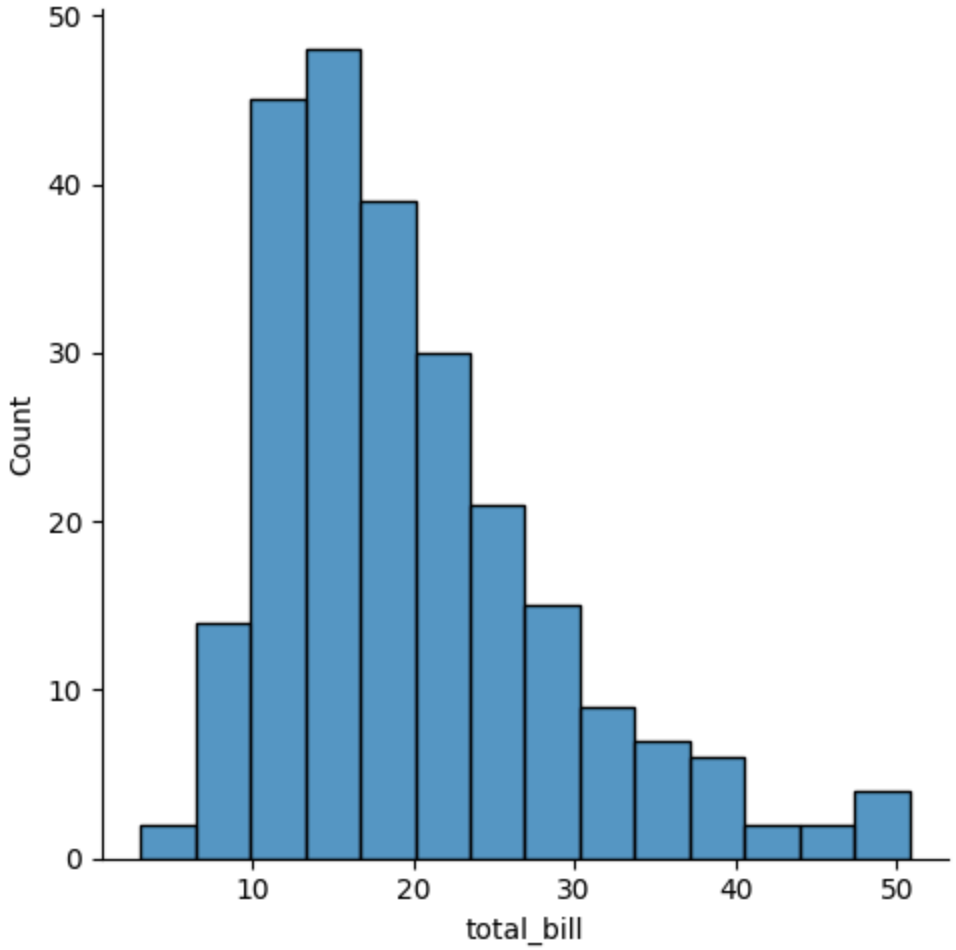

sns.displot(tips.total_bill,kde=False)
plt.show()
sns.jointplot(x=tips.tip,y=tips.total_bill)
plt.show()
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")plt.show
()
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
plt.show()
```

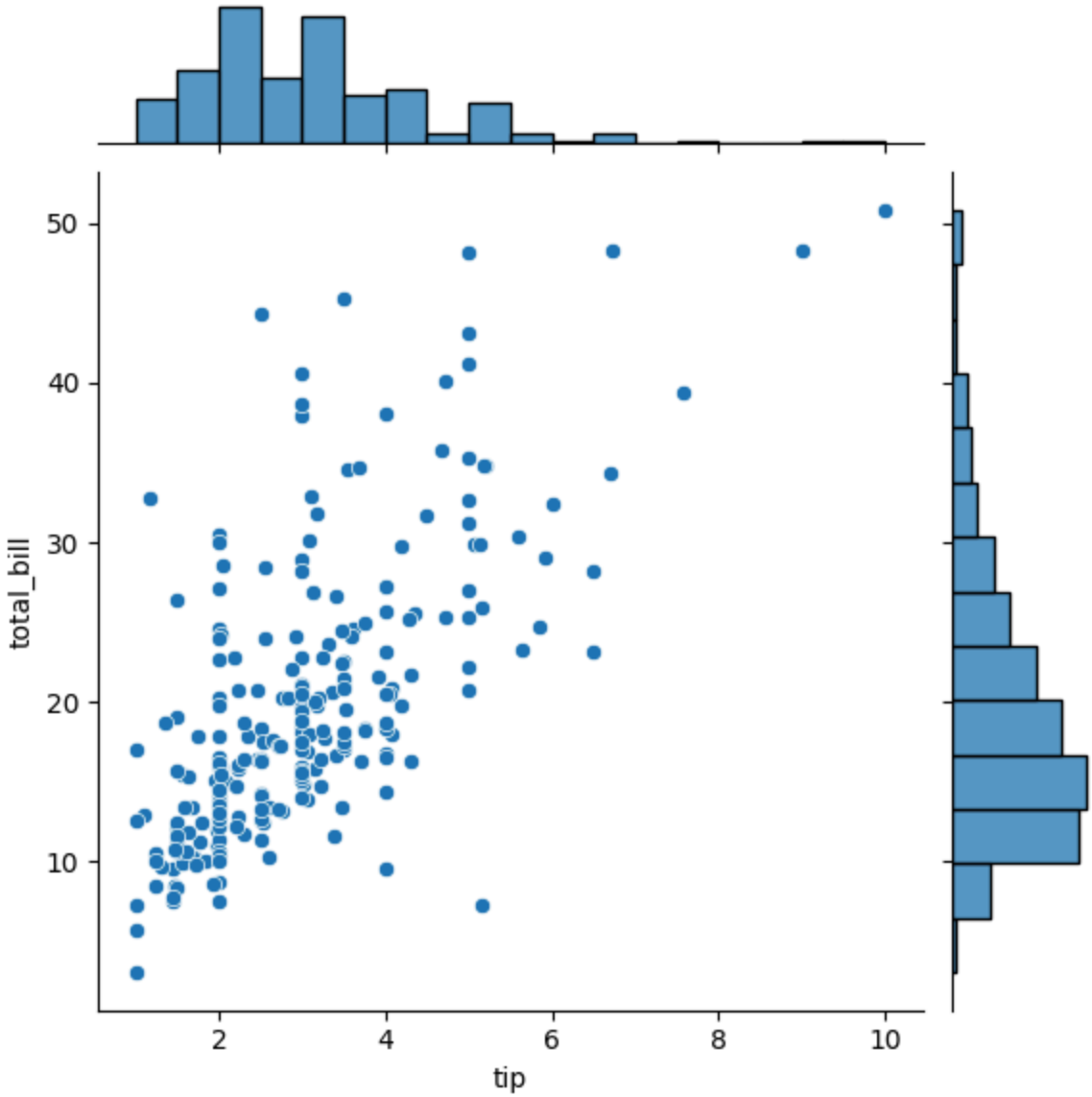
```
sns.pairplot(tips)
plt.show()
print(tips.time.value_counts())
print("\n\n")
sns.pairplot(tips,hue='time')
plt.show()
sns.pairplot(tips,hue='day')
plt.show()
sns.heatmap(tips.corr(numeric_only=True),annot=True)
plt.show()
sns.boxplot(tips.total_bill)
plt.show()
sns.boxplot(tips.tip)
plt.show()
sns.countplot(tips.day)
plt.show()
sns.countplot(tips.sex)
plt.show()
tips.sex.value_counts().plot(kind='pie')
plt.show()
tips.sex.value_counts().plot(kind='bar')
plt.show()
sns.countplot(tips[tips.time=='Dinner']['day'])
plt.show()
```

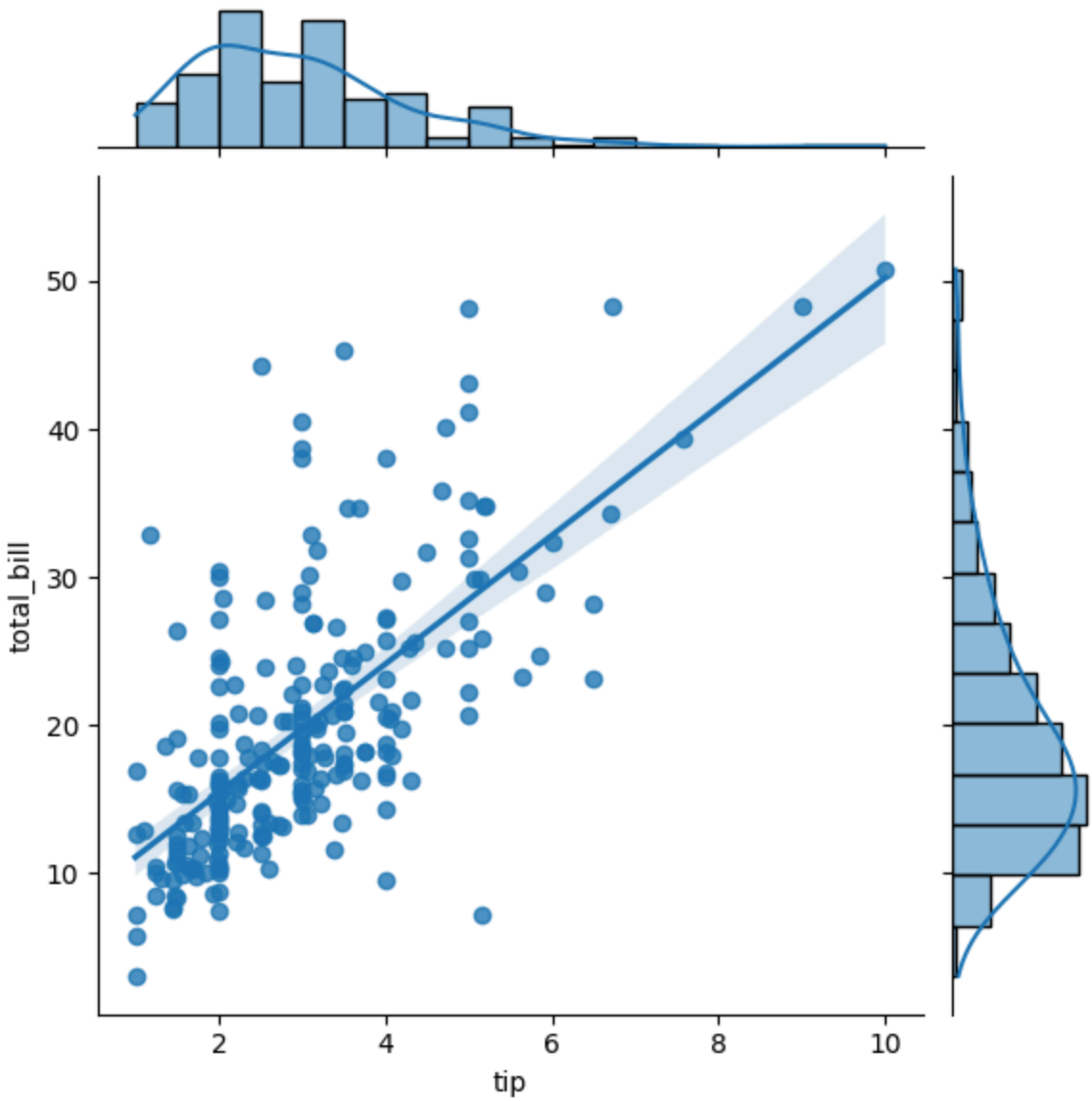
OUTPUT:

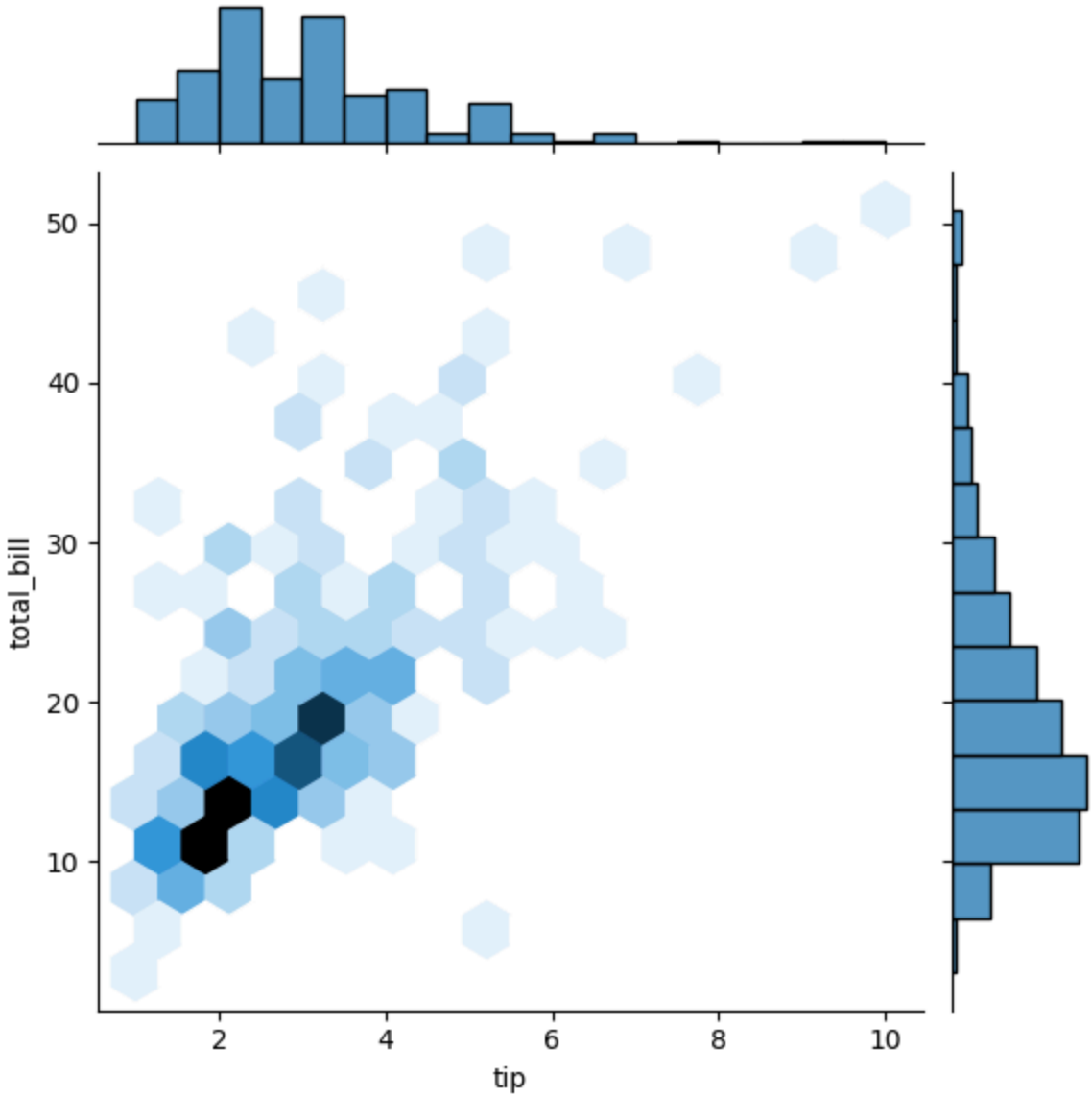
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

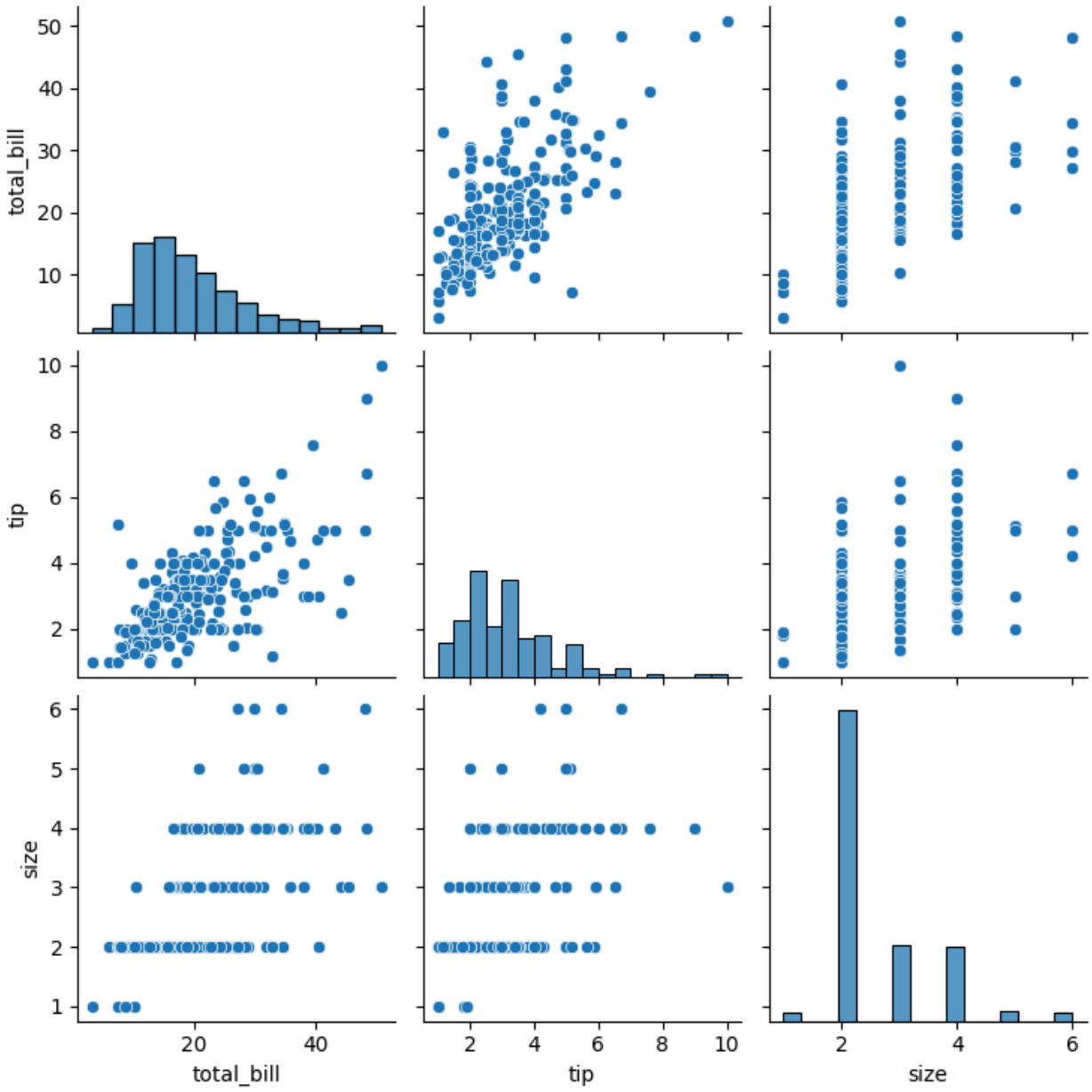

50
40
30
20
10
0
Count
10
20
30
40
50
total_bill


50
40
30
20
10
0
Count
10
20
30
40
50
total_bill

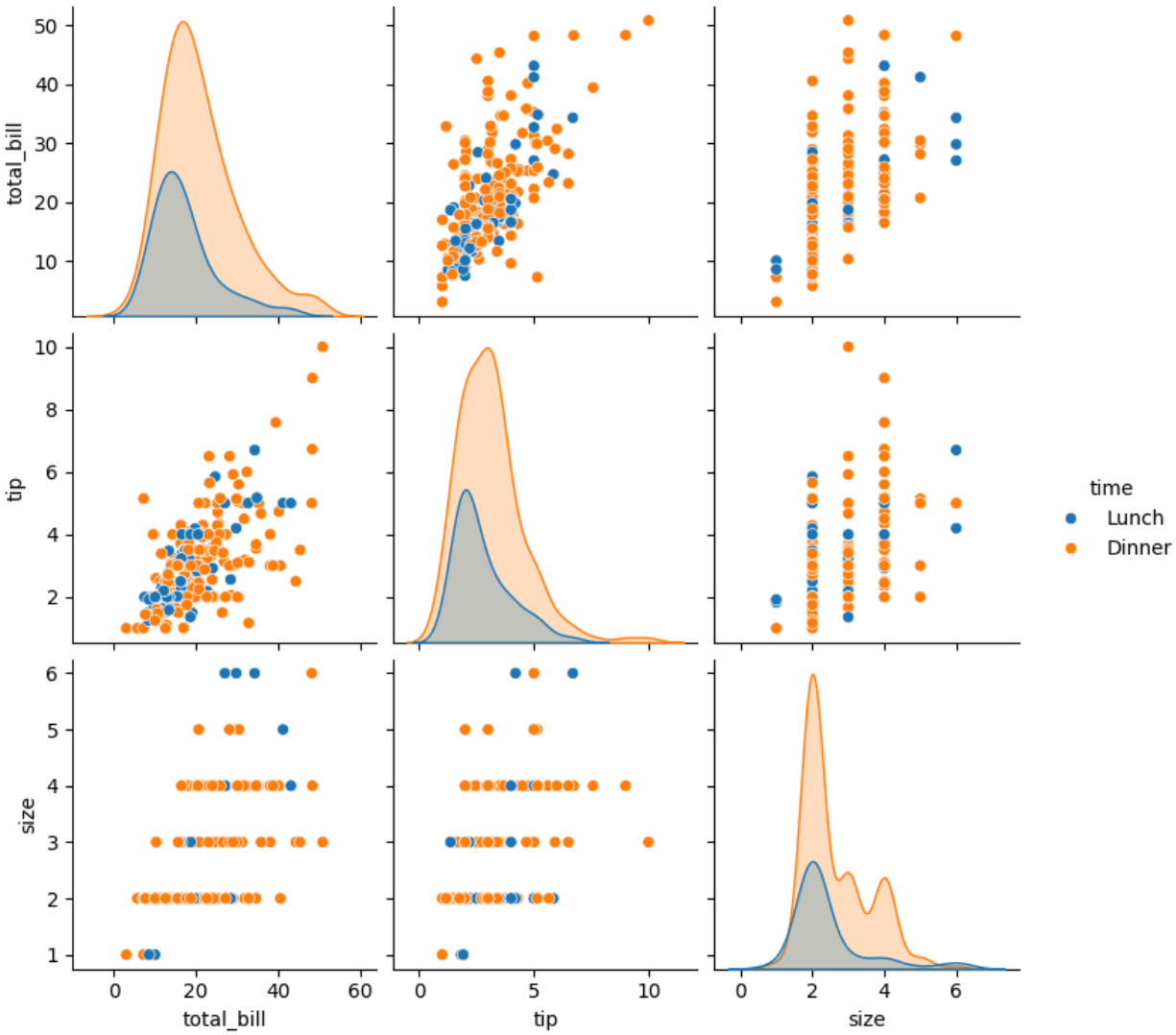

50
40
30
20
10
total_bill
2
4
6
8
10
tip

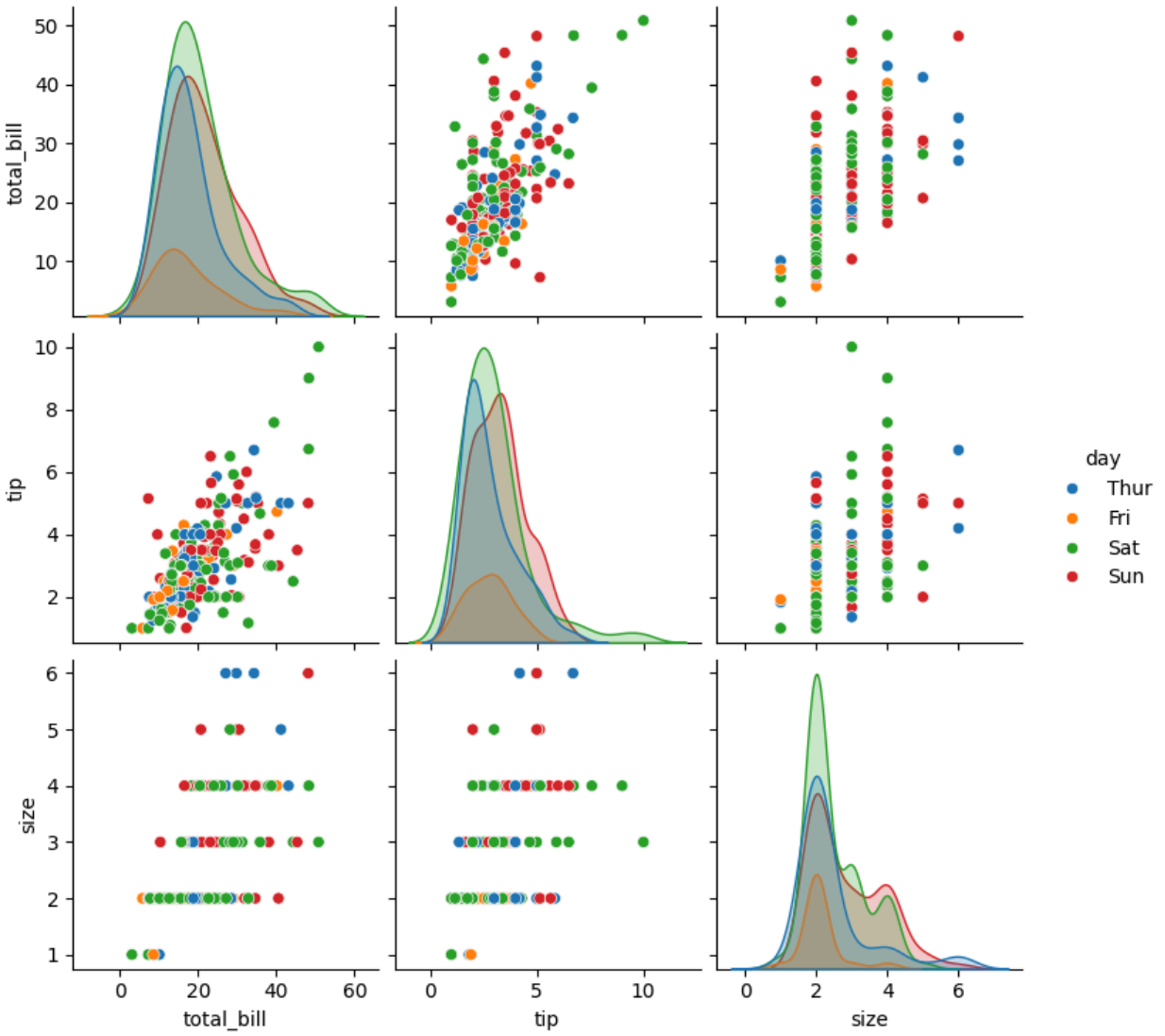

total_bill
tip
50
40
30
20
10
2
4
6
8
10

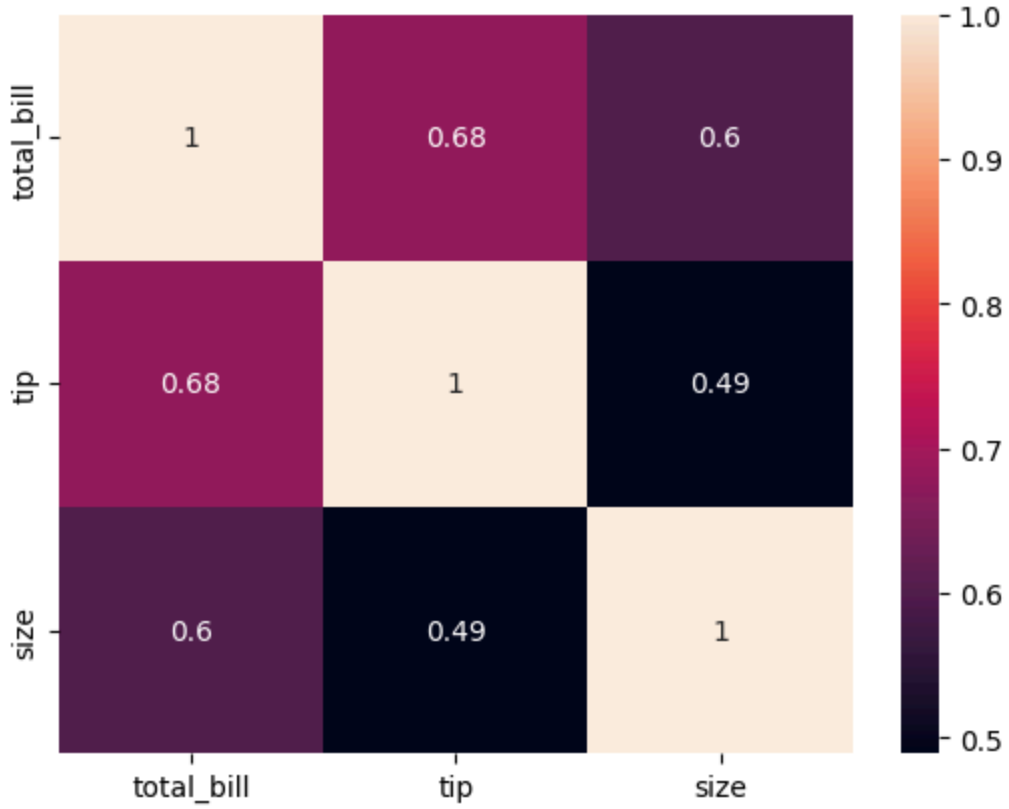

50
40
30
20
10
total_bill
2
4
6
8
10
tip

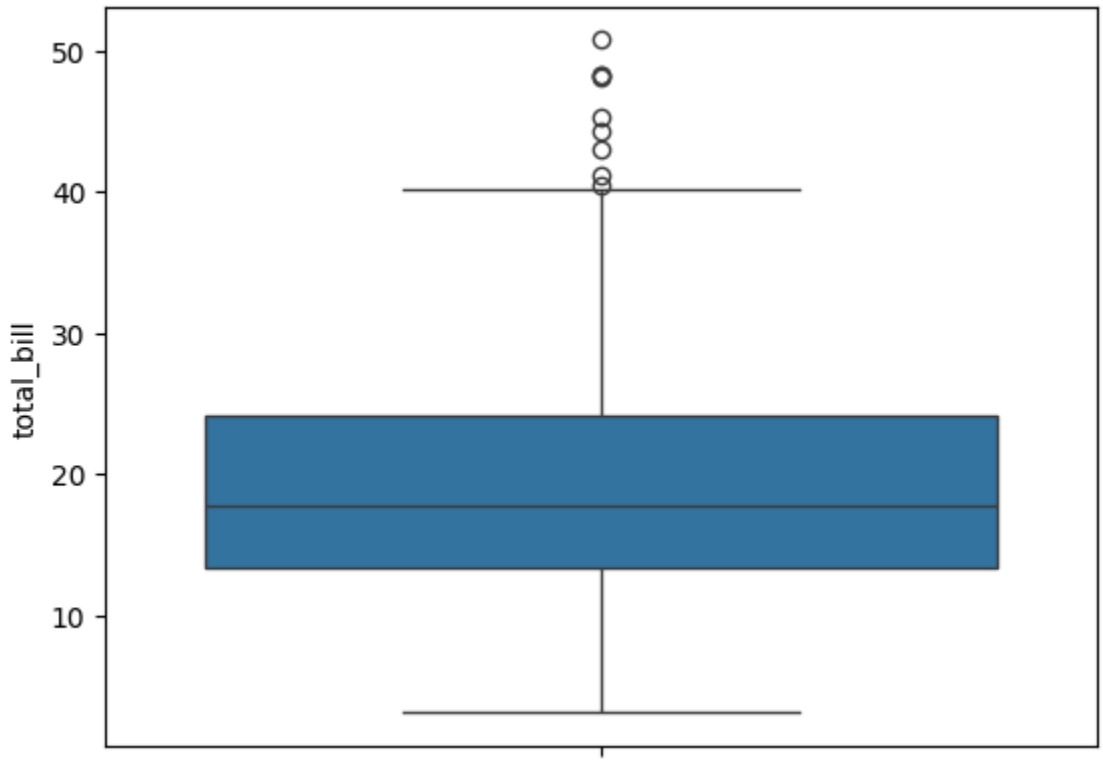


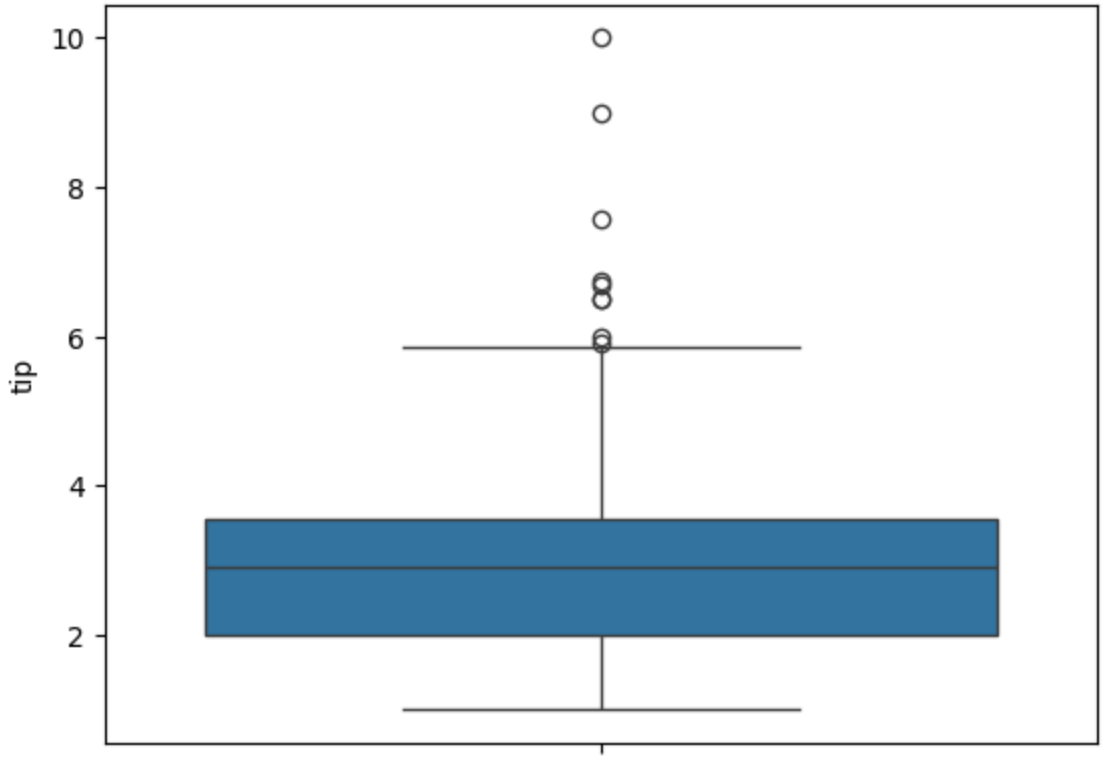
time
Dinner 176
Lunch 68
Name: count, dtype: int64

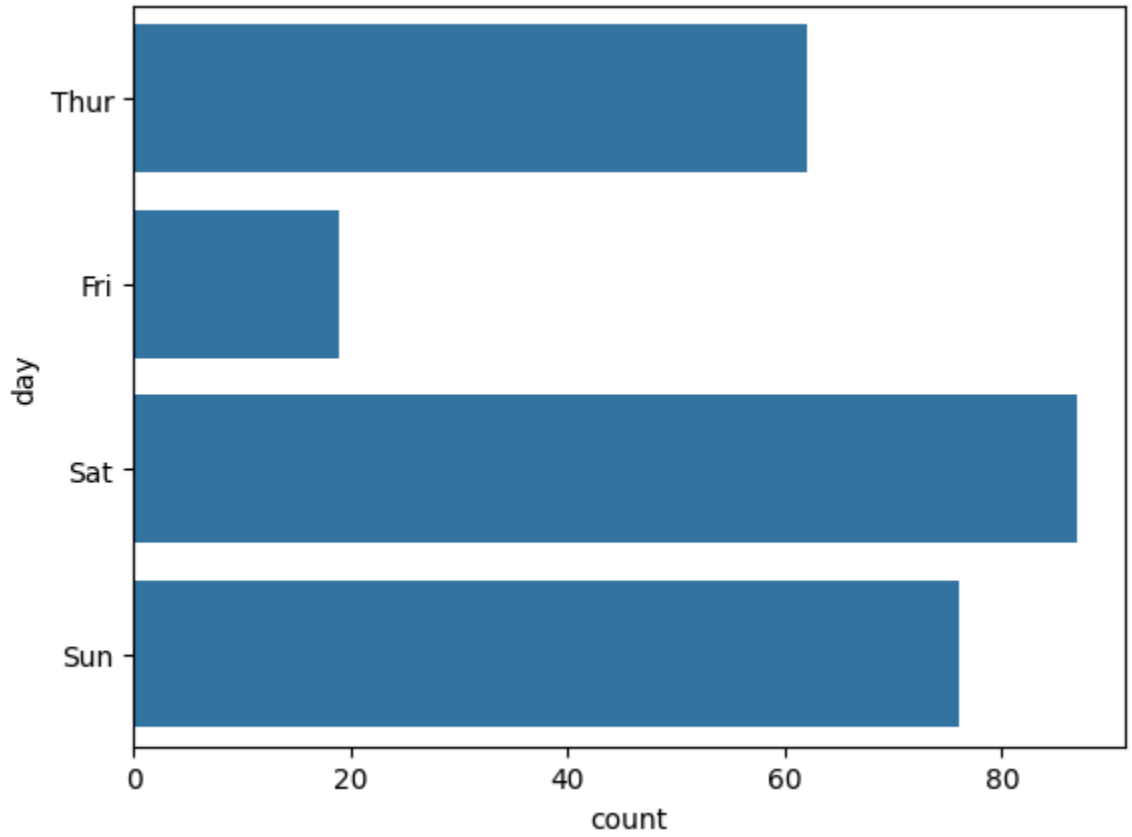

total_bill
tip
size
time
Lunch
Dinner

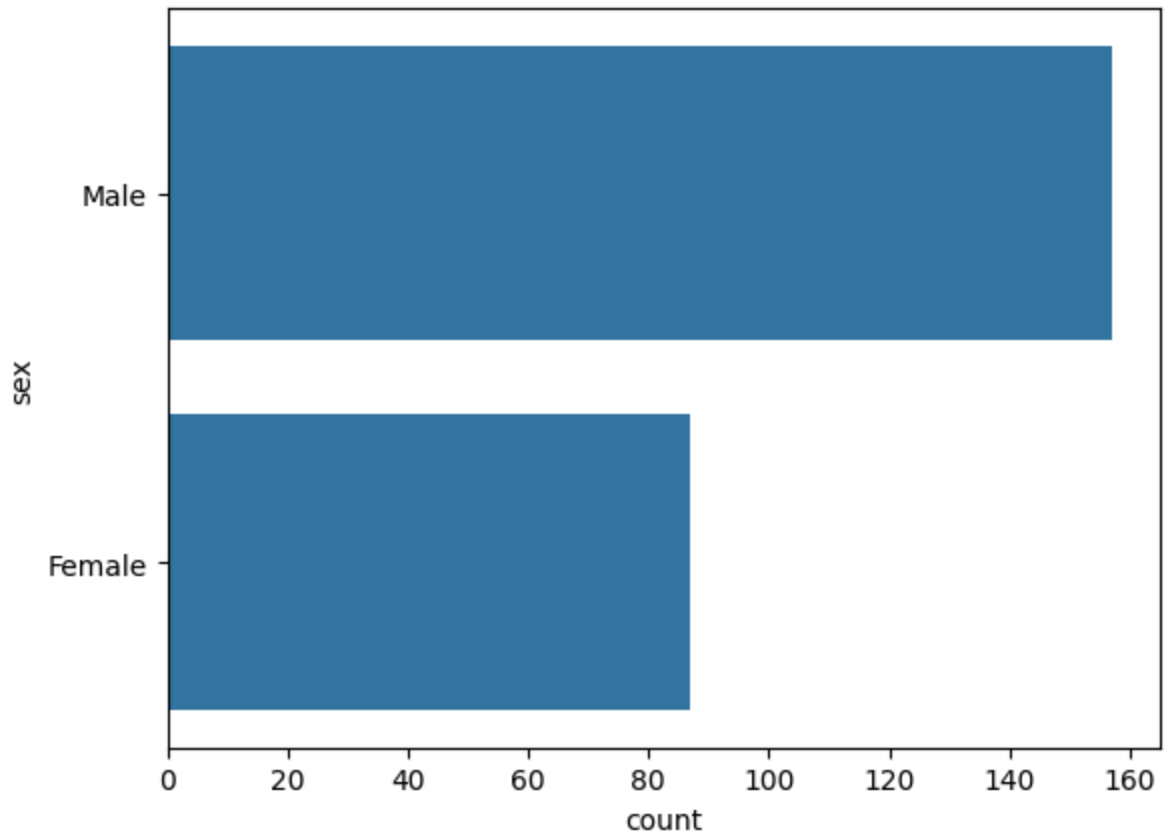

total_bill
tip
size
day
Thur
Fri
Sat
Sun

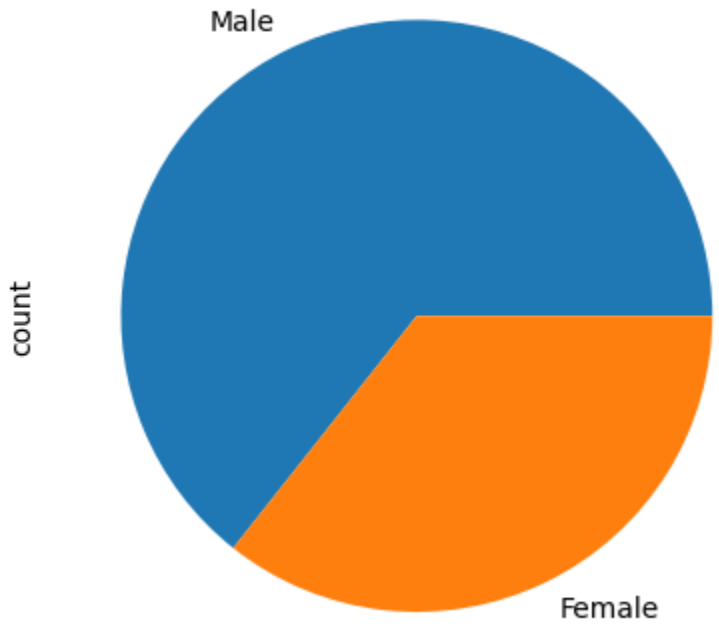

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

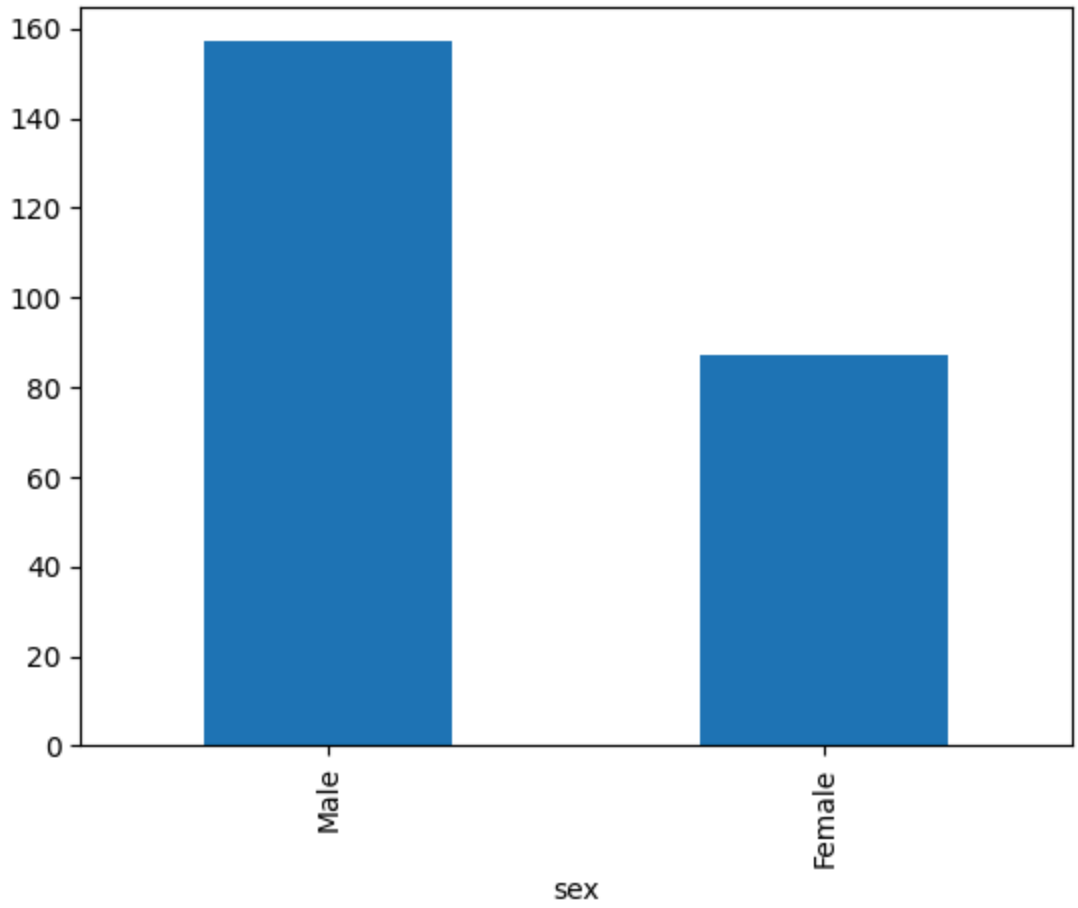

total_bill
50
40
30
20
10

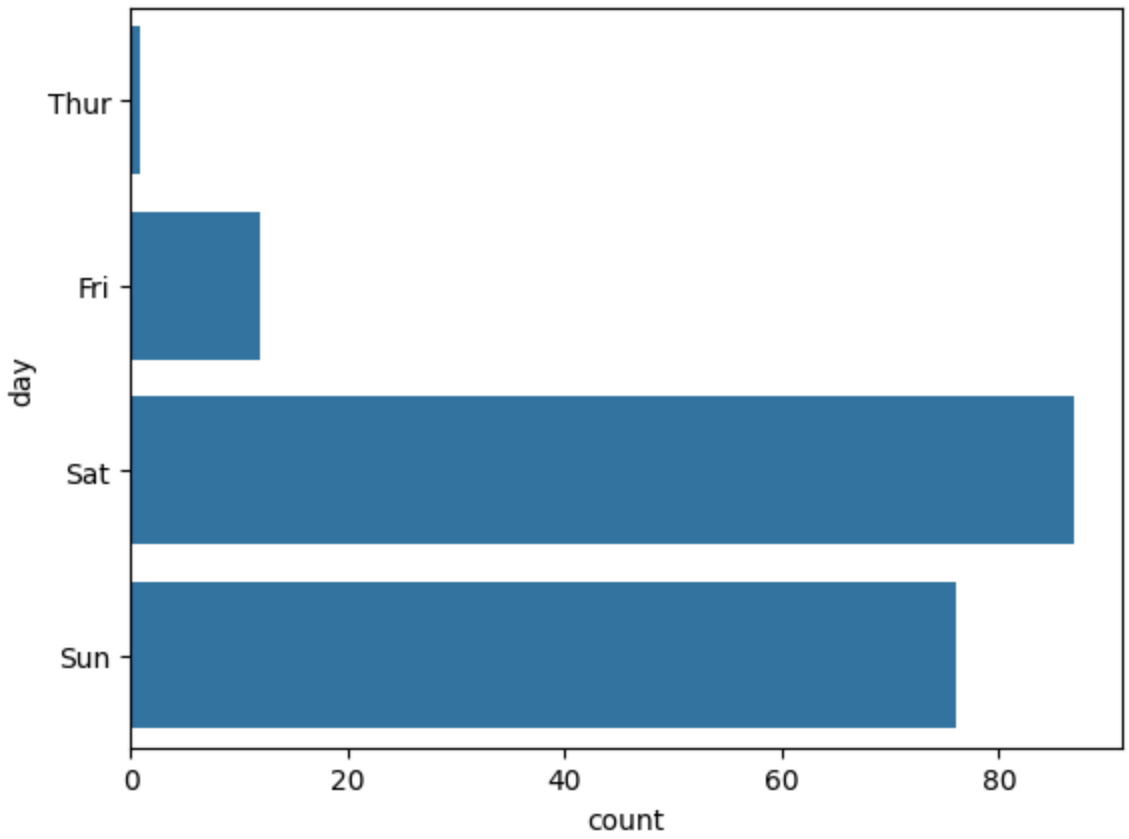

10
8
6
4
2
tip


Thur
Fri
Sat
Sun
day
0
20
40
60
80
count


Male
Female
sex
0
20
40
60
80
100
120
140
160
count


Male
Female
count


160
140
120
100
80
60
40
20
0
Male
Female
sex



RESULT:

Thus, Exploratory Data Analysis (EDA) was successfully performed on the Seaborn `tips` dataset using quantitative and qualitative analysis. Quantitative analysis was carried out using distribution plots, joint plots, pair plots, box plots, and a correlation heatmap. Qualitative analysis was performed using count plots, bar charts, and pie charts to study categorical variables such as `sex`, `day`, and `time`. The dataset was analyzed to understand distributions, relationships, and patterns in the data.